Maanav Anand Kumar

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Education

Stony Brook University, ***Master of Science in Computer Science*** **Aug. 2024 – May 2026**
- **GPA:** 3.73/4 New York, USA
- **Relevant Coursework:** Analysis of Algorithms, Operating Systems, Wireless Networks, Hybrid Systems

Chennai Institute of Technology, ***Bachelor of Engineering in Computer Science*** **Aug. 2020 – May 2024**
- **GPA:** 8.91/10 Chennai, India
- **Relevant Coursework:** Distributed Systems, Data Structures and Algorithms, Computer Networks, Software Engineering, Object Oriented Programming, Database Management Systems, Cloud Computing, Machine Learning

Experience

New York Wealth Planning Group, Software Engineer Intern **New York, USA | May 2025 – Aug. 2025**
- Deployed full-stack web applications using React.js, Node.js, and MySQL on AWS EC2, architected to handle over 10K concurrent sessions. Implemented CI/CD pipelines using GitHub Actions and Docker, ensuring seamless scalability.
- Designed and maintained RESTful backend services handling 150K+ monthly requests, improving reliability through Redis caching, optimized MySQL queries, connection pooling, and HTTP performance tuning, reducing API latency by 22%.

CGI Information Systems, SDE Apprentice (ML Systems) **Chennai, India | May 2024 – July 2024**
- Developed and optimized machine learning models using Python, TensorFlow, and scikit-learn, integrating them into containerized services with Docker and Kubernetes to enhance automation and scalability of agile predictive analytics modules.
- Refactored ML service components to optimize memory usage and data I/O, reducing latency by $\approx$15%; added unit and regression tests to improve reliability of distributed deployments.

Timescan Logistics, Software Developer Intern **Chennai, India | Jun. 2023 – Dec. 2023**
- Engineered automation scripts and data integration to synchronize multi-branch logistics operations, reducing manual workload by 35% and improving turnaround time for trans-continental shipment processing, contributing to measurable business growth.
- Maintained enterprise logistics applications using Flask, MySQL, and REST APIs; optimized SQL queries and database access patterns to improve reliability of internal production services.

Projects

Distributed Service Monitoring Platform **|Personal Project**
- Developed a production-grade monitoring service in Go that tracks website health, response latency, and uptime metrics through REST APIs, structured JSON logging, and thread-safe in-memory storage. Leveraged goroutines, channels, and mutexes to concurrently monitor multiple services while maintaining data consistency under parallel workloads.
- Deployed the platform on AWS EC2 (Ubuntu Linux) with Nginx reverse proxy and systemd service management, enabling automatic recovery, graceful shutdown handling, and continuous background health checks. Implemented rolling historical metrics and live reporting endpoints to support operational visibility and service reliability analysis.

Web Based Telemetry and Control Dashboard **|Interacting Robotic Systems Lab, New York**
- Built a modular web-based telemetry dashboard enabling real-time visualization and remote control of IoT and robotic devices using a microservice architecture with Flask, WebSockets, and ROSBridge, ensuring reliable data exchange across distributed nodes.
- Designed backend pipelines for concurrent sensor streams, achieving sub-200 ms message latency through asynchronous I/O, optimized serialization, and Redis-backed message caching. Deployed on AWS EC2 instances (Linux) using Docker.
- Implemented health monitoring and diagnostic tooling for distributed robotic nodes, reducing troubleshooting effort and improving observability during system deployment and testing.

Software Framework for ML Training and Evaluation Platform **|Stony Brook University, New York**
- Developed a scalable deep reinforcement learning (DRL) framework in Python using TensorFlow and PyTorch to simulate and evaluate autonomous navigation policies for high-speed, collision-free driving in virtual F1TENTH track environments.
- Implemented a robust API layer and configuration management system to automate experiment setup, parameter tuning, and performance tracking across model variants, supporting seamless integration with data visualization dashboards.

Skills

Programming Languages Python, Go, C++, Java, JavaScript, SQL
Systems Linux/Unix, Bash, TCP/IP Networking, Debugging, Distributed Systems, Observability
Backend & Databases REST APIs, Flask, Node.js, MySQL, PostgreSQL, MongoDB, Redis
Cloud & Infrastructure AWS EC2, Docker, Kubernetes, Nginx, CI/CD, GitHub Actions

Open Source Contributions

F1TENTH Trajectory-Aided Learning (TAL) Controller **GitHub**
- Contributed to an autonomous racing framework for the F1TENTH platform, implementing and evaluating reinforcement learning–based control policies for high-speed vehicle navigation. Improved experimentation workflows and controller evaluation pipelines, supporting reproducible benchmarking of learning-based navigation algorithms.

Ubuntu, PyCloudLib and PythiaLabs **GitHub**
- Contributed documentation updates, bug fixes, and tests to OSS projects, improving usability and developer experience